

SEAN MAROOFI

Research Assistant, Hamburg University of Technology, Germany

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RESEARCH INTERESTS

Robotics, Control, Deep Learning, Dynamic & Clustered Environments

EDUCATION

Hamburg University of Technology, Hamburg, Germany

October 2020 — August 2023

Master of Science in Mechatronics

- Minor: Robotics and Intelligent Systems
- Thesis: Dynamic Model Inversion of a Soft Robot

University of California, Berkeley, California, USA

August 2021 — May 2022

Study Abroad, Faculty of Mechanical Engineering

- Focus: Advanced Control, Machine Learning, Deep Learning
- Project: Reinforcement Learning Interface for Autonomous Driving

Hamburg University of Technology, Hamburg, Germany

October 2017 — September 2020

Bachelor of Science in Mechanical Engineering

- Minor: Mechatronics
- Thesis: Acoustic Localization of Micro Underwater Robots by Means of Kalman Filtering

RESEARCH EXPERIENCE

Hamburg University of Technology

Hamburg, Germany

Research Assistant

June 2024 — Present

Affiliation: Institute of Logistics Engineering

- Image-based prediction of localization loss of mobile robots in warehouse environments.
- Multi-modal robot dataset (MRCD) for large-scale outdoor SLAM.
- Development of machine learning framework (Flowcean) for cyber-physical systems.
- Lead maintainer and developer of a state-machine architecture for indoor mobile robot control and navigation.
- Navigation of mobile delivery robots in public space in compliance with Germany VDA.
- Supervision of master thesis on semantic scene classification for robot navigation in pedestrian areas.
- Acquisition of third-party funding.

Hamburg University of Technology

Hamburg, Germany

Master Student Researcher

January 2023 — September 2023

Affiliation: Institute of Mechanics and Ocean Engineering

Advisors: Prof. Robert Seifried, Prof. Christian J. Cyron

- Feedforward model-based control of a beam-shaped soft robot for dynamic trajectory tracking.
- Addressing forward dynamics inversion problem by derivation of inverse dynamics using the servo-constraints approach.
- Analysis of numerical stability, invertibility, and controllability.
- Experimental evaluation of trajectory tracking with a self-built robot test system.

University of California, Berkeley

Hamburg, Germany

Graduate Student Researcher

August 2021 — May 2022

Affiliation: Mechanical Systems Control Laboratory

Advisors: Prof. Masayoshi Tomizuka, Prof. Robert Seifried

- Design of a framework for reinforcement learning (RL) control for autonomous driving using the INTERACTION dataset.
- Application of soft-actor-critic for learning a driving policy in dynamic, dense urban traffic environment.
- Optimization of the framework and simulation with respect to suitability as a training environment and user experience.

WORKING EXPERIENCE

Altona Analytics GmbH

March 2024 — May 2024

Automation Engineer

- Automation of in-vitro PCR test evaluation.
- Programming of robot workstation for dilution and pipetting processes.

Optima Pressformen GmbH & Co. KG

September 2017 — October 2017

Technical Internship

- Construction of a hand-made pneumatic-driven machine using metalworking techniques.
- Incoming and outgoing good inspection of shafts and metal parts.

ACADEMIC PROJECTS

Flowcean

Hamburg, Germany

Institute of Logistics Engineering

December 2024 — Present

- A cross-domain framework for automatic generation of machine learning models for cyber-physical systems.
- Design of ROS2 Interface to deploy models in robotic use cases.

TaBuLa-LOG+

Hamburg, Germany

Institute of Logistics Engineering

June 2024 — December 2024

- Integration of automated postal delivery robots in public transport.
- Development of software stack for robot control and navigation.

Deep Learning Student Project

California, USA

Graduate course CS282, Berkeley

January 2022 — May 2022

- Using vision-transformers for feature extraction of contextual semantic images of F1 racing.
- Combination with a trajectory transformer to predict future trajectories of multiple race cars.

Medical Robotics Student Project

Hamburg, Germany

Master course on robotics and navigation in medicine

April 2021 — August 2021

- Kinematic control of a 7-axis robot manipulator for medical needle injection into a human thorax.

PUBLICATIONS

Reviewed

- [1] M. Knitt*, **S. Maroofi***, M. Thakkar*, H. Rose, P. Braun, "Synthetic Datasets for Data-Driven Localization Monitoring," *Logistics Journal*, 2025

Submitted

- [1] **S. Maroofi**, M. Grube, R. Seifried, "Dynamic Model Inversion of a Soft Robot using Servo Constraints," *under review*, 2025
- [2] J. Ziegenbein*, N. Blunder*, **S. Maroofi***, M. Thiel, T. Nguyen, H. Rose, P. Braun, C. Jahn, "Mobile Robot Campus Dataset for Evaluating SLAM Algorithms on Wheeled Robots," *under review*, 2025

Workshop

- [1] M. Knitt*, **S. Maroofi***, M. Schmidt, M. Thakkar, "Creating Data-Driven Models for Robot Localization Systems using Flowcean," *International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, 2025
- [2] M. Thiel, N. Blunder, J. Ziegenbein, **S. Maroofi**, H. Rose, "Integrating Delivery Robots into Public Transport," *Workshop on Autonomous Delivery and Service Robots on Pedestrian and Cycle Paths*, 2025

(The superscript* indicates equal contribution)

AWARDS AND SCHOLARSHIPS

International Study and Training Partnerships (ISAP)

Hamburg, Germany

DAAD Scholarship for study abroad at University of California, Berkeley

2021

ENGLISH

TOEFL iBT: 105 (overall score)

Listening: 25 — Reading: 23

Speaking: 29 — Writing: 28

Test date: December 2024

SKILLS

- **Programming:** Python, C++, ROS, Matlab, Java, R
- **Software:** NVIDIA Isaac Sim, Git, Docker, Linux, LaTeX, Inkscape, Gimp, Office
- **Languages:** German, English, Latin, Spanish

REFERENCES

Prof. Robert Seifried

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Scholar Profiles: Hamburg University of Technology - Personal Page — Google Scholar — LinkedIn

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Philipp Braun

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